OMDENA · TERRAYIELD ANALYTICS PHASE II

Crop Yield Benchmarking

Features, Baselines, and Evaluation

Sprint 3 · Workshop 1



TODAY'S LEARNING PATH

What we will cover

01	**Frame the benchmark**	task, scope, scientific question
02	**Define the evidence**	targets, labels, dataset unit
03	**Build valid features**	Sentinel-2, climate, soil, management
04	**Prepare and split**	quality checks, leakage, spatial/temporal tests
05	**Compare models**	mean, linear, random forest, boosting
06	**Evaluate results**	metrics, model comparison, error analysis
07	**Wrap up and questions**	key takeaways, Q&A

Sprint 3 turns prepared data into comparable ML evidence

Governed data products

Known inputs
Documented preprocessing
Reusable geospatial assets

Measurable ML evidence

Comparable runs
Defensible evaluation
Reusable benchmark artifacts

Shared benchmark decisions are the bridge

Without a common task, label, split and metric, model scores cannot be compared—or trusted in Sprint 4.

BENCHMARK SCOPE

Start with two questions—not with a model

PRIMARY TASK

Crop-yield regression

Question How accurately can we predict harvested yield for an unseen field-season?
Output Continuous yield in tonnes per hectare (t/ha)
Unit One agricultural parcel × one growing season
Decision use Regional forecasting and risk monitoring

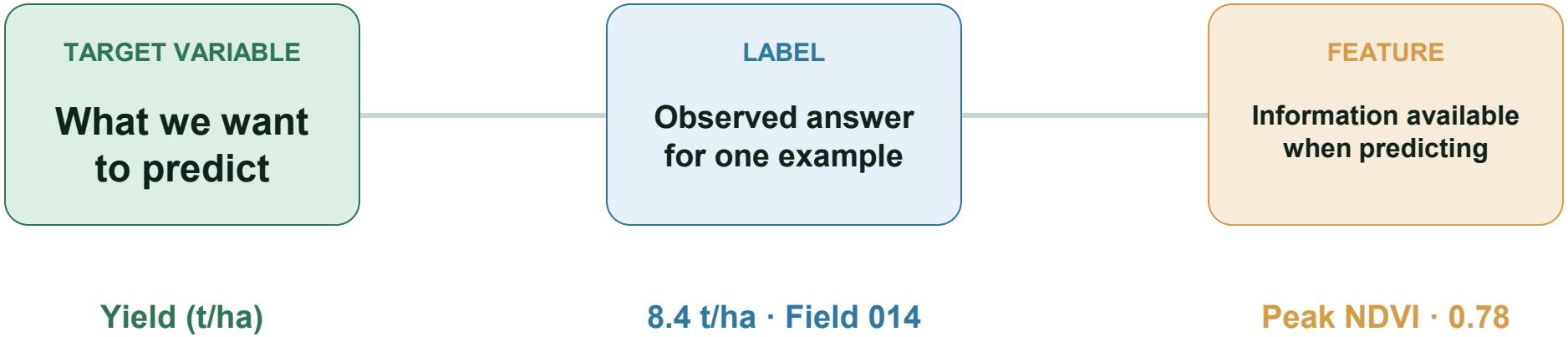
COMPANION TASK

Land-use change classification

Question Did a location change land-cover class between two defined dates?
Output Class / change label, defined before training
Unit Pixel, patch or parcel—one must be chosen
Decision use Change screening and map review

Today’s depth: yield regression. Today’s alignment: both task definitions and their shared data-governance rules.

Target, label and feature answer different questions



Scientific validity starts here: a feature must be available before the forecast cut-off; a label must represent the intended outcome.

One benchmark row should mean one field in one season

field_id	season	crop	peak_ndvi	mean_ndmi	yield_t_ha
F014	2025	winter wheat	0.78	0.31	8.4

The target contract

Definition: harvested crop mass ÷ harvested area
Unit: tonnes per hectare (t/ha)
Time key: crop season and harvest year
Source: measured or audited harvest record—not satellite imagery

Reject or quarantine

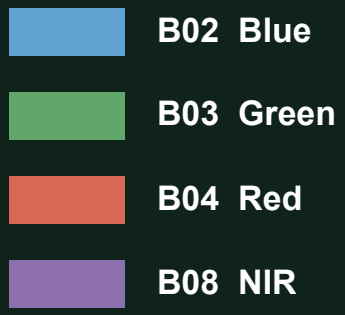
Missing field or season key
Mixed crop definitions
Inconsistent mass / area units
Impossible or unexplained yield values

REMOTE-SENSING INPUTS

Sentinel-2 observes the canopy—not the harvested yield

What the sensor sees

Surface reflectance across spectral bands



What we derive

NDVI — vegetation vigor
NDMI — canopy moisture
SWIR — moisture / structure signal
Time series — green-up, peak and senescence

Derived features need cloud masking, common grids and consistent dates.

What it cannot prove

Actual harvested mass
Farm management intent
Causal effect of one input
Performance in a new region

Satellite variables are predictive evidence—not ground truth.

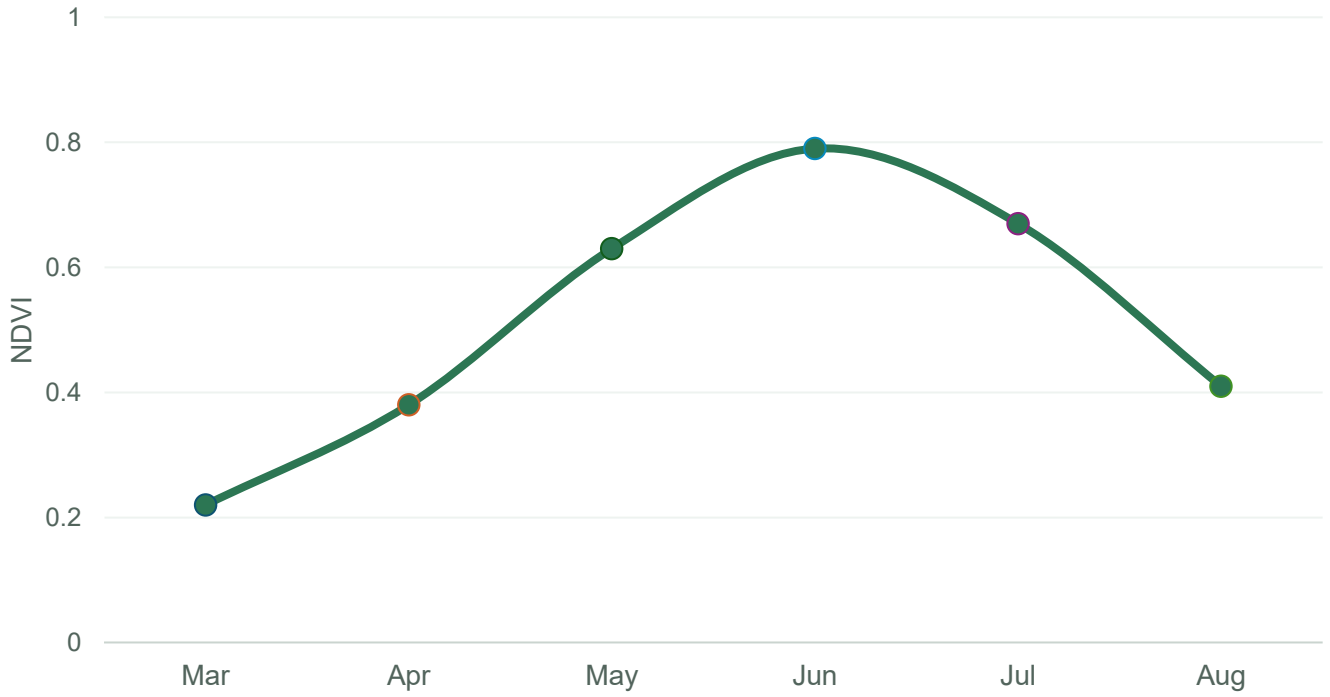
FEATURE CATEGORIES

Yield features must capture crop, context and time

Climate	rainfall · temperature · radiation
Soil	texture · pH · organic matter
Management	crop type · sowing · irrigation
Remote sensing	NDVI · NDMI · SWIR · red edge
Spatial & temporal	region · elevation · day-of-year

Validity rule: every feature needs a source, timestamp, spatial unit and forecast-time availability check.

The season matters more than one green snapshot



Raw inputs

Six cloud-screened observations

Engineered features

Peak NDVI = 0.79
Peak timing = June
Green-up slope
Senescence slope
Cumulative NDVI

Domain knowledge compresses noisy measurements into interpretable signals about crop development.

Preprocessing is part of the scientific method

1
Audit
missing · duplicates · units

2
Mask
cloud · shadow · no-data

3
Align
coordinate system · grid · resolution

4
Aggregate
pixel → parcel → season

5
Transform
encode · scale · engineer

Three non-negotiable checks

Fit on training data only: missing-value and scaling settings must come from the training data.
Use the right mask for each task: do not remove a valid target class by reusing the wrong mask.
Field-level quality: scene cloud percentage does not guarantee a usable field observation.

Leakage gives the model information it should not have

Looks good—fails later

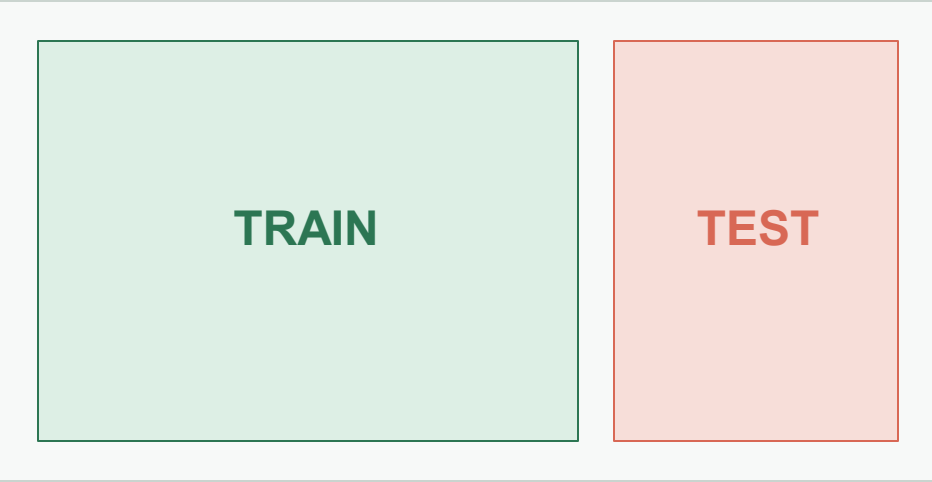
Randomly mix nearby fields



Train and test share the same local weather, soil and neighborhood signal.

Harder—but honest

Keep an entire region or later season separate



This separate test set represents a real deployment gap.

Future information is also leakage: post-harvest weather, later imagery, or labels embedded in features.

A held-out test set stays unseen until final evaluation

Held out = kept separate while models are trained and compared; used only for the final evaluation.

Split	What it tests	Use when	Main risk
Random	new rows from a similar distribution	non-spatial, independent rows	optimistic with nearby fields
Temporal	a future season	forecasting later years	one unusual year dominates
Spatial	an unseen region	geographic transfer	region imbalance
Spatial + temporal	new place and future season	deployment evidence	needs more data

Whichever split is chosen, every model must be evaluated on the same untouched test set.

Start simple, then test whether complexity adds value

01

Mean predictor

Reference baseline: predict the training-set average for every field.

02

Linear regression

Interpretable baseline: test whether a simple relationship—such as peak NDVI—already adds value.

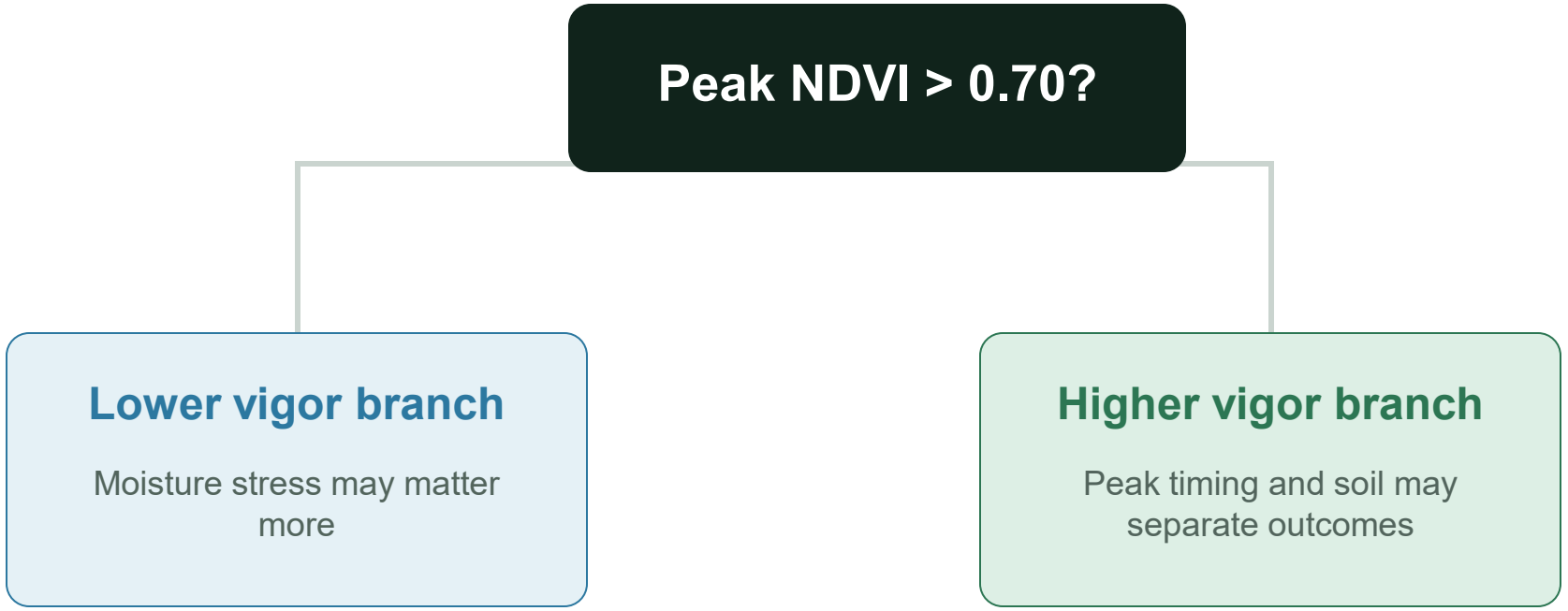
03

Tree ensemble

Candidate model: test whether nonlinear interactions across vigor, moisture and timing improve prediction.

Rule: use the same features, split and regression metrics for every model.

Tree regressors capture thresholds and interactions



Why ensembles?

Random Forest
Averages many trees
Gradient Boosting
Adds trees to correct earlier errors
Both
Capture nonlinear patterns

Limit tree depth, choose settings on validation data, and treat importance as association—not causation.

Follow one field through the benchmark

1

Field-season

F014 · winter wheat · 2025

2

Allowed features

peak NDVI 0.78 · NDMI 0.31 · peak day 171

3

Predictions

mean 7.6 · linear 8.1 · forest 8.3 t/ha

4

Observed label

8.4 t/ha after harvest

Forest absolute error = |8.3 − 8.4| = 0.1 t/ha

The label is revealed only after prediction; the test row never trains the model.

EVALUATION METRICS

Metrics follow the prediction task—not the model

Mean, linear, forest and boosting models all predict continuous yield, so they use the same regression metrics.

PRIMARY TASK

Crop-yield regression

MAE · primary	Typical prediction error in t/ha; use it to rank yield models.
RMSE · supporting	Gives more weight to large misses, where severe errors matter.
R^2 · supporting	Shows improvement relative to the mean baseline; it is unitless.

Same metrics for every crop-yield model.

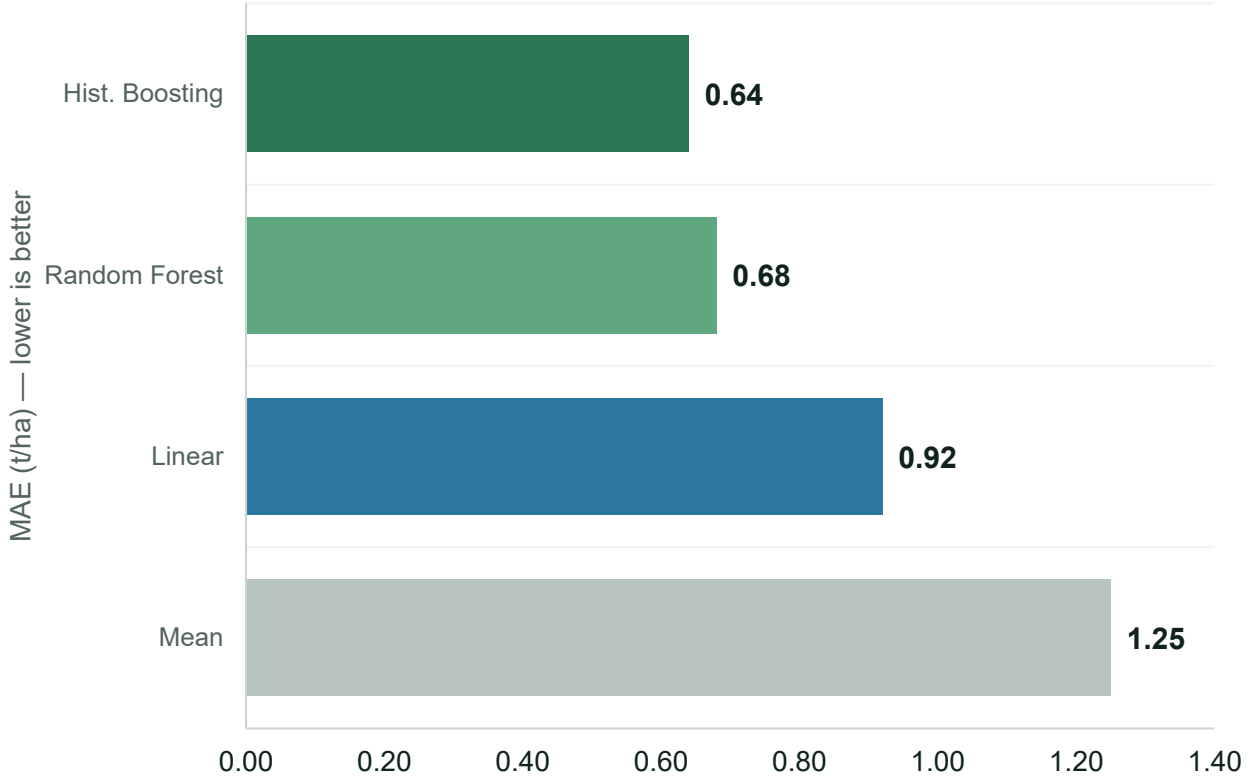
COMPANION TASK

Land-use change classification

Macro F1 · primary
Balances precision and recall for each class.
Balanced accuracy · supporting
Accounts for unequal class sizes.
Confusion matrix
Shows which classes are being mixed up.

The task determines the metric; the model does not.

Compare every yield model on the same held-out test set



ILLUSTRATIVE · NOT PROJECT RESULTS

What the comparison suggests

Every model uses the same held-out test rows
All learned models beat the mean reference
Boosting’s edge over forest is small

Error analysis before choosing a model

MAE by held-out region and crop
Residuals vs. yield and cloud coverage
Large-error fields and data-quality flags

KEY TAKEAWAYS

A credible benchmark keeps the rules consistent

01	**Clear task**	one prediction unit, crop, geography and forecast date
02	**Trusted label**	measured or audited yield, correct unit and season alignment
03	**Valid features**	inputs available at forecast time; satellite signals are predictors
04	**Honest test**	a held-out region or season, untouched while models are chosen
05	**Fair comparison**	the same split and metrics for every baseline and model

When these rules are documented and reused, benchmark results remain comparable and defensible.

Any questions?

Thank you

Crop Yield Benchmarking · TerraYield Analytics Phase II